

Praktik JavaScript (JS)

🌸 A. Bahan Bacaan

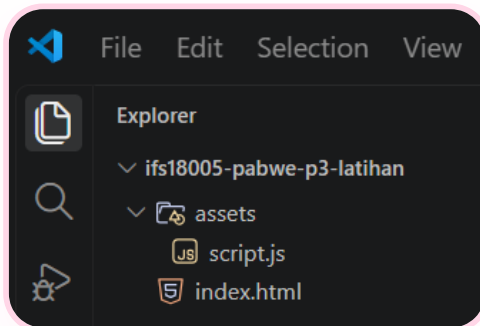
🌻 1.1 JavaScript (JS)

Pelajari JavaScript dari halaman berikut:

<https://www.w3schools.com/js/default.asp>

🌸 B. Latihan

1. Buat struktur proyek seperti berikut:



2. Modifikasi isi file `assets/script.js`, seperti berikut:

🐾 Coding Time

```
1  /**
2   * Latihan JavaScript – Praktikum 3
3   * Fitur: Tab switcher, Todo (CRUD + localStorage), Magic Number, Password Checker
4   */
5
6  /* ===== UTILITAS ===== */
7
8  /** Ambil elemen; lempar error jika tidak ada (membantu debug DOM) */
9  function $(selector) {
10     const el = document.querySelector(selector);
11     if (!el) throw new Error(`Elemen tidak ditemukan: ${selector}`);
12     return el;
13 }
14
15 function $all(selector) {
16     return document.querySelectorAll(selector);
17 }
18
19 /* ===== TAB SWITCHER ===== */
20
21 const TAB_STORAGE_KEY = "pabwe-p3-active-tab";
22 const tabButtons = $all(".tab-btn");
23 const panels = {
24     todo: $("#panel-todo"),
25     magic: $("#panel-magic"),
```

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26   password: $("#panel-password"),
27   };
28
29   /** Ganti tab aktif: sembunyikan panel lain, highlight tombol, ingat di localStorage */
30   function switchTab(name) {
31     if (!panels[name]) name = "todo";
32
33     Object.entries(panels).forEach(([key, panel]) => {
34       panel.classList.toggle("hidden", key !== name);
35     });
36
37     tabButtons.forEach((btn) => {
38       const active = btn.dataset.tab === name;
39       btn.setAttribute("aria-selected", String(active));
40       btn.classList.toggle("bg-sky-600", active);
41       btn.classList.toggle("text-white", active);
42       btn.classList.toggle("shadow", active);
43       btn.classList.toggle("text-slate-600", !active);
44       btn.classList.toggle("hover:bg-slate-100", !active);
45     });
46
47     localStorage.setItem(TAB_STORAGE_KEY, name);
48   }
49
50   tabButtons.forEach((btn) => {
51     btn.addEventListener("click", () => switchTab(btn.dataset.tab));
52   });
53
54   // Pulihkan tab terakhir yang dibuka (default: todo)
55   const savedTab = localStorage.getItem(TAB_STORAGE_KEY) || "todo";
56   switchTab(savedTab);
57
58   /* ===== TODO LIST ===== */
59
60   const TODO_STORAGE_KEY = "pabwe-p3-todos";
61   const STATUS_ORDER = ["Todo", "Progress", "Done", "Cancel"];
62
63   /** Warna badge status */
64   const STATUS_STYLE = {
65     Todo: "bg-sky-100 text-sky-800",
66     Progress: "bg-amber-100 text-amber-800",
67     Done: "bg-emerald-100 text-emerald-800",
68     Cancel: "bg-rose-100 text-rose-800",
69   };
70
71   let todos = loadTodos();
72   let editingId = null; // id todo yang sedang diubah di modal
73   let deletingId = null; // id todo yang akan dihapus di modal
74
75   const todoForm = $("#todo-form");
76   const todoTitle = $("#todo-title");
77   const todoStatus = $("#todo-status");
78   const todoSearch = $("#todo-search");
79   const todoSort = $("#todo-sort");
80   const todoList = $("#todo-list");
81   const todoEmpty = $("#todo-empty");
82
83   const modalEdit = $("#modal-edit");
84   const modalDelete = $("#modal-delete");
85   const editForm = $("#edit-form");
86   const editTitle = $("#edit-title");

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87 | const editStatus = $("#edit-status");
88 | const deleteTodoTitle = $("#delete-todo-title");
89 | const deleteConfirm = $("#delete-confirm");
90 |
91 | /** Baca data dari localStorage */
92 | function loadTodos() {
93 |   try {
94 |     const raw = localStorage.getItem(TODO_STORAGE_KEY);
95 |     return raw ? JSON.parse(raw) : [];
96 |   } catch {
97 |     return [];
98 |   }
99 | }
100 |
101 | /** Simpan data ke localStorage */
102 | function saveTodos() {
103 |   localStorage.setItem(TODO_STORAGE_KEY, JSON.stringify(todos));
104 | }
105 |
106 | /** Filter + sort daftar, lalu render ke DOM */
107 | function renderTodos() {
108 |   const query = todoSearch.value.trim().toLowerCase();
109 |   const sort = todoSort.value;
110 |
111 |   let items = todos.filter((t) => t.title.toLowerCase().includes(query));
112 |
113 |   items = [...items].sort((a, b) => {
114 |     switch (sort) {
115 |       case "oldest":
116 |         return a.createdAt - b.createdAt;
117 |       case "title-asc":
118 |         return a.title.localeCompare(b.title, "id");
119 |       case "title-desc":
120 |         return b.title.localeCompare(a.title, "id");
121 |       case "status":
122 |         return STATUS_ORDER.indexOf(a.status) - STATUS_ORDER.indexOf(b.status);
123 |       case "newest":
124 |       default:
125 |         return b.createdAt - a.createdAt;
126 |     }
127 |   });
128 |
129 |   // Alert jika belum ada todo sama sekali
130 |   const noTodos = todos.length === 0;
131 |   todoEmpty.classList.toggle("hidden", !noTodos);
132 |   todoList.classList.toggle("hidden", noTodos);
133 |
134 |   if (noTodos) {
135 |     todoList.innerHTML = "";
136 |     return;
137 |   }
138 |
139 |   // Kosongkan lalu bangun ulang lewat DOM
140 |   todoList.innerHTML = "";
141 |
142 |   if (items.length === 0) {
143 |     const li = document.createElement("li");
144 |     li.className =
145 |       "rounded-xl border border-slate-200 bg-slate-50 px-4 py-3 text-sm text-slate-600";
146 |     li.textContent = "Tidak ada todo yang cocok dengan pencarian.";
147 |     todoList.appendChild(li);

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148     return;
149 }
150
151 items.forEach((todo) => {
152     const li = document.createElement("li");
153     li.className =
154         "flex flex-col sm:flex-row sm:items-center gap-3 rounded-xl border border-slate-200 px-4
py-3";
155     li.dataset.id = todo.id;
156
157     const info = document.createElement("div");
158     info.className = "flex-1 min-w-0";
159
160     const titleEl = document.createElement("p");
161     titleEl.className = "font-medium text-slate-900 truncate";
162     titleEl.textContent = todo.title;
163
164     const badge = document.createElement("span");
165     badge.className = `inline-flex mt-1 text-xs font-semibold px-2 py-0.5 rounded-md
${STATUS_STYLE[todo.status] || ""}`;
166     badge.textContent = todo.status;
167
168     info.append(titleEl, badge);
169
170     const actions = document.createElement("div");
171     actions.className = "flex items-center gap-1.5 shrink-0";
172
173     // Tombol ubah → buka modal edit
174     const editBtn = document.createElement("button");
175     editBtn.type = "button";
176     editBtn.className =
177         "inline-flex items-center gap-1 rounded-lg border border-slate-200 px-2.5 py-1.5 text-xs
font-semibold text-slate-700 hover:bg-slate-50";
178     editBtn.innerHTML = '<i class="ti ti-pencil"></i> Ubah';
179     editBtn.addEventListener("click", () => openEditModal(todo.id));
180
181     // Tombol hapus → buka modal konfirmasi
182     const deleteBtn = document.createElement("button");
183     deleteBtn.type = "button";
184     deleteBtn.className =
185         "inline-flex items-center gap-1 rounded-lg border border-rose-200 px-2.5 py-1.5 text-xs
font-semibold text-rose-700 hover:bg-rose-50";
186     deleteBtn.innerHTML = '<i class="ti ti-trash"></i> Hapus';
187     deleteBtn.addEventListener("click", () => openDeleteModal(todo.id));
188
189     actions.append(editBtn, deleteBtn);
190     li.append(info, actions);
191     todoList.appendChild(li);
192 });
193 }
194
195 /** Tampilkan / sembunyikan modal (pakai flex agar center) */
196 function openModal(modal) {
197     modal.classList.remove("hidden");
198     modal.classList.add("flex");
199     document.body.classList.add("overflow-hidden");
200 }
201
202 function closeModal(modal) {
203     modal.classList.add("hidden");
204     modal.classList.remove("flex");

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205     document.body.classList.remove("overflow-hidden");
206 }
207
208 /** Buka modal ubah: isi field dengan data todo */
209 function openEditModal(id) {
210     const todo = todos.find((t) => t.id === id);
211     if (!todo) return;
212
213     editingId = id;
214     editTitle.value = todo.title;
215     editStatus.value = todo.status;
216     openModal(modalEdit);
217     editTitle.focus();
218 }
219
220 /** Buka modal hapus: tampilkan judul todo yang akan dihapus */
221 function openDeleteModal(id) {
222     const todo = todos.find((t) => t.id === id);
223     if (!todo) return;
224
225     deletingId = id;
226     deleteTodoTitle.textContent = `${todo.title}`;
227     openModal(modalDelete);
228 }
229
230 /** Tutup modal ubah */
231 function closeEditModal() {
232     editingId = null;
233     editForm.reset();
234     closeModal(modalEdit);
235 }
236
237 /** Tutup modal hapus */
238 function closeDeleteModal() {
239     deletingId = null;
240     closeModal(modalDelete);
241 }
242
243 // Tutup lewat tombol Batal / x / klik backdrop
244 $all("[data-close-modal='edit']").forEach((btn) => {
245     btn.addEventListener("click", closeEditModal);
246 });
247 $all("[data-close-modal='delete']").forEach((btn) => {
248     btn.addEventListener("click", closeDeleteModal);
249 });
250 modalEdit.querySelector(".modal-backdrop").addEventListener("click", closeEditModal);
251 modalDelete.querySelector(".modal-backdrop").addEventListener("click", closeDeleteModal);
252
253 // Escape menutup modal yang terbuka
254 document.addEventListener("keydown", (e) => {
255     if (e.key !== "Escape") return;
256     if (!modalEdit.classList.contains("hidden")) closeEditModal();
257     if (!modalDelete.classList.contains("hidden")) closeDeleteModal();
258 });
259
260 /** Simpan perubahan dari modal edit */
261 editForm.addEventListener("submit", (e) => {
262     e.preventDefault();
263     const title = editTitle.value.trim();
264     if (!title || !editingId) return;
265

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266   const todo = todos.find((t) => t.id === editingId);
267   if (todo) {
268     todo.title = title;
269     todo.status = editStatus.value;
270     saveTodos();
271     renderTodos();
272   }
273
274   closeEditModal();
275 });
276
277 /** Konfirmasi hapus dari modal */
278 deleteConfirm.addEventListener("click", () => {
279   if (!deletingId) return;
280
281   todos = todos.filter((t) => t.id !== deletingId);
282   saveTodos();
283   renderTodos();
284   closeDeleteModal();
285 });
286
287 /** Form utama: hanya untuk menambah todo baru */
288 todoForm.addEventListener("submit", (e) => {
289   e.preventDefault();
290   const title = todoTitle.value.trim();
291   const status = todoStatus.value;
292   if (!title) return;
293
294   todos.push({
295     id: crypto.randomUUID(),
296     title,
297     status,
298     createdAt: Date.now(),
299   });
300
301   saveTodos();
302   todoForm.reset();
303   todoStatus.value = "Todo";
304   renderTodos();
305 });
306
307 todoSearch.addEventListener("input", renderTodos);
308 todoSort.addEventListener("change", renderTodos);
309
310 renderTodos();
311
312 /* ===== MAGIC NUMBER ===== */
313
314 const MAGIC_TOP_KEY = "pabwe-p3-magic-top";
315
316 let secretNumber = 0;
317 let attempts = 0;
318 let gameOver = false;
319
320 const magicForm = $("#magic-form");
321 const magicGuess = $("#magic-guess");
322 const magicAttempts = $("#magic-attempts");
323 const magicTopScore = $("#magic-top-score");
324 const magicFeedback = $("#magic-feedback");
325 const magicReset = $("#magic-reset");
326

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327  /** Ambil top score (jumlah tebakan terbaik = terkecil) */
328  function getTopScore() {
329      const v = localStorage.getItem(MAGIC_TOP_KEY);
330      return v ? Number(v) : null;
331  }
332
333  function showTopScore() {
334      const top = getTopScore();
335      magicTopScore.textContent = top === null ? "-" : `${top} tebakan`;
336  }
337
338  /** Tampilkan pesan feedback ke pemain */
339  function setMagicFeedback(message, type = "info") {
340      const styles = {
341          info: "border-slate-200 bg-slate-50 text-slate-700",
342          high: "border-orange-200 bg-orange-50 text-orange-900",
343          low: "border-sky-200 bg-sky-50 text-sky-900",
344          win: "border-lime-200 bg-lime-50 text-lime-900",
345      };
346      const icons = {
347          info: "ti-info-circle",
348          high: "ti-arrow-up",
349          low: "ti-arrow-down",
350          win: "ti-trophy",
351      };
352
353      magicFeedback.className = `rounded-xl border px-4 py-3 text-sm flex items-start gap-2
354      ${styles[type]}`;
355      magicFeedback.innerHTML = `

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386   if (!isWholeNumberInput(raw)) {
387       setMagicFeedback("Tebakan harus bilangan bulat (contoh: 42), bukan desimal.", "info");
388       magicGuess.focus();
389       return;
390   }
391
392   const guess = Number(raw);
393   if (guess < 1 || guess > 1000) {
394       setMagicFeedback("Masukkan bilangan bulat antara 1 sampai 1000.", "info");
395       magicGuess.focus();
396       return;
397   }
398
399   attempts += 1;
400   magicAttempts.textContent = String(attempts);
401
402   if (guess > secretNumber) {
403       // Tebakan lebih besar dari angka rahasia
404       setMagicFeedback(`Tebakan ${guess} terlalu besar. Coba angka lebih kecil.`, "high");
405   } else if (guess < secretNumber) {
406       // Tebakan lebih kecil dari angka rahasia
407       setMagicFeedback(`Tebakan ${guess} terlalu kecil. Coba angka lebih besar.`, "low");
408   } else {
409       // Benar!
410       gameOver = true;
411       magicGuess.disabled = true;
412
413       const top = getTopScore();
414       let msg = `Benar! Angkanya ${secretNumber}. Kamu butuh ${attempts} tebak.`;
415
416       // Update top score jika lebih baik (lebih sedikit tebak)
417       if (top === null || attempts < top) {
418           localStorage.setItem(MAGIC_TOP_KEY, String(attempts));
419           showTopScore();
420           msg += " Rekor baru!";
421       }
422
423       setMagicFeedback(msg, "win");
424   }
425
426   magicGuess.value = "";
427   magicGuess.focus();
428 });
429
430 magicReset.addEventListener("click", startMagicGame);
431 startMagicGame();
432
433 /* ===== PASSWORD CHECKER ===== */
434 /* Penilaian mengikuti praktik modern (NIST/OWASP-inspired):
435  * panjang lebih penting dari sekadar komposisi; password umum & pola lemah dihukum. */
436
437 const passwordInput = $("#password-input");
438 const passwordToggle = $("#password-toggle");
439 const strengthBar = $("#password-strength-bar");
440 const strengthLabel = $("#password-strength-label");
441 const strengthHint = $("#password-strength-hint");
442
443 /** Password yang sering bocor / mudah ditebak (contoh edukatif) */
444 const COMMON_PASSWORDS = new Set([
445     "password", "password1", "password123", "123456", "12345678", "123456789",
446     "1234567890", "qwerty", "qwerty123", "abc123", "admin", "admin123",

```

```

447 "welcome", "welcome1", "iloveyou", "letmein", "monkey", "dragon",
448 "master", "login", "passwd", "p@ssw0rd", "p@ssword", "changeme",
449 "indonesia", "jakarta", "sayang", "rahasia", "password!",
450 ];
451
452 const SEQUENCES = [
453   "0123456789",
454   "abcdefghijklmnopqrstuvwxyz",
455   "qwertyuiop",
456   "asdfghjkl",
457   "zxcvbnm",
458 ];
459
460 /** kriteria checklist UI */
461 const passwordRules = {
462   length: (pw) => pw.length >= 12, // rekomendasi modern (OWASP sering ≥12)
463   upper: (pw) => /[A-Z]/.test(pw),
464   lower: (pw) => /[a-z]/.test(pw),
465   number: (pw) => /[0-9]/.test(pw),
466   special: (pw) => /[!@#%&*~.-_]/.test(pw), // sembarang non-alfanumerik
467   noCommon: (pw) => pw.length > 0 && !hasWeakPattern(pw),
468 };
469
470 const STRENGTH_META = [
471   { label: "Belum diisi", bar: "w-0 bg-slate-300", text: "text-slate-500", hint: "" },
472   {
473     label: "Sangat lemah",
474     bar: "w-1/5 bg-rose-500",
475     text: "text-rose-600",
476     hint: "Mudah ditebak atau terlalu pendek. Hindari password umum.",
477   },
478   {
479     label: "Lemah",
480     bar: "w-2/5 bg-orange-500",
481     text: "text-orange-600",
482     hint: "Masih rentan. Perpanjang dan hindari pola berurutan.",
483   },
484   {
485     label: "Cukup",
486     bar: "w-3/5 bg-amber-500",
487     text: "text-amber-600",
488     hint: "Layak untuk akun biasa, tapi masih bisa diperkuat.",
489   },
490   {
491     label: "Kuat",
492     bar: "w-4/5 bg-lime-500",
493     text: "text-lime-700",
494     hint: "Sudah baik: panjang dan beragam karakter.",
495   },
496   {
497     label: "Sangat kuat",
498     bar: "w-full bg-emerald-500",
499     text: "text-emerald-700",
500     hint: "Sesuai praktik modern untuk akun penting.",
501   },
502 ];
503
504 /** Deteksi urutan keyboard / alfabet / angka */
505 function hasSequence(pw) {
506   const lower = pw.toLowerCase();
507   for (const seq of SEQUENCES) {

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508     for (let i = 0; i <= seq.length - 4; i++) {
509         const chunk = seq.slice(i, i + 4);
510         const rev = [...chunk].reverse().join("");
511         if (lower.includes(chunk) || lower.includes(rev)) return true;
512     }
513 }
514 return false;
515 }
516
517 /** Karakter sama berulang ≥3 kali (aaa, 111) */
518 function hasRepeatedChars(pw) {
519     return /(.)\1{2,}/.test(pw);
520 }
521
522 /** Password umum atau pola lemah */
523 function hasweakPattern(pw) {
524     const lower = pw.toLowerCase();
525     if (COMMON_PASSWORDS.has(lower)) return true;
526     // Variasi sederhana: password123!, qwerty!
527     for (const common of COMMON_PASSWORDS) {
528         if (lower.includes(common) && common.length >= 5) return true;
529     }
530     if (hasSequence(pw) || hasRepeatedChars(pw)) return true;
531     return false;
532 }
533
534 /** Estimasi ukuran charset (untung diversity / entropi kasar) */
535 function charsetSize(pw) {
536     let size = 0;
537     if (/[a-z]/.test(pw)) size += 26;
538     if (/[A-Z]/.test(pw)) size += 26;
539     if (/[0-9]/.test(pw)) size += 10;
540     if (/[^A-Za-z0-9]/.test(pw)) size += 33; // perkiraan simbol umum
541     return size;
542 }
543
544 /**
545  * Skor kekuatan 0-5 (bukan sekadar jumlah centang).
546  * Panjang & entropi diutamakan; pola lemah menurunkan skor.
547  */
548 function scorePasswordStrength(pw) {
549     if (!pw) return 0;
550
551     const len = pw.length;
552     const pool = charsetSize(pw);
553     // Entropi kasar: log2(pool^length) ≈ length * log2(pool)
554     const entropy = pool > 1 ? len * Math.log2(pool) : 0;
555
556     let score = 1; // mulai dari sangat lemah jika ada isi
557
558     // Panjang (faktor utama standar modern)
559     if (len >= 8) score = 2;
560     if (len >= 12) score = 3;
561     if (len >= 16) score = 4;
562     if (len >= 20) score = 5;
563
564     // Bonus keragaman charset (minimal 3 kelas karakter)
565     const classes =
566         Number(/[a-z]/.test(pw)) +
567         Number(/[A-Z]/.test(pw)) +
568         Number(/[0-9]/.test(pw)) +

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```

569     Number(/^A-Za-z0-9]/.test(pw));
570     if (classes >= 3 && score < 5) score += 1;
571     if (classes >= 4 && len >= 12 && score < 5) score += 1;
572
573     // Naikkan/turunkan berdasarkan entropi
574     if (entropy < 28) score = Math.min(score, 1);
575     else if (entropy < 36) score = Math.min(score, 2);
576     else if (entropy < 50) score = Math.min(score, 3);
577     else if (entropy < 70) score = Math.min(score, 4);
578
579     // Penalti keras untuk password umum / pola lemah
580     if (hasWeakPattern(pw)) score = Math.min(score, 1);
581     if (len < 8) score = Math.min(score, 1); // di bawah batas NIST minimum
582
583     return Math.max(0, Math.min(5, score));
584 }
585
586 /** Update centang kriteria + progress bar kekuatan */
587 function checkPassword() {
588     const pw = passwordInput.value;
589
590     Object.entries(passwordRules).forEach(([rule, test]) => {
591         const ok = test(pw);
592         const item = document.querySelector(`.criterion[data-rule="${rule}"]`);
593         if (!item) return;
594
595         const icon = item.querySelector("i");
596         item.classList.toggle("text-emerald-700", ok);
597         item.classList.toggle("text-slate-500", !ok);
598         icon.className = `ti ${ok ? "ti-circle-check-filled text-emerald-600" : "ti-circle"} text-
base shrink-0`;
599     });
600
601     const score = scorePasswordStrength(pw);
602     const meta = STRENGTH_META[score];
603
604     strengthBar.className = `h-full rounded-full transition-all duration-300 ${meta.bar}`;
605     strengthLabel.className = `font-semibold ${meta.text}`;
606     strengthLabel.textContent = meta.label;
607     strengthHint.textContent = meta.hint;
608 }
609
610 passwordInput.addEventListener("input", checkPassword);
611
612 /** Toggle tampil / sembunyikan password */
613 passwordToggle.addEventListener("click", () => {
614     const show = passwordInput.type === "password";
615     passwordInput.type = show ? "text" : "password";
616     passwordToggle.innerHTML = `

```

3. Modifikasi isi file `index.html`, seperti berikut:

 Coding Time

```

1 <!DOCTYPE html>
2 <html lang="id">
3 <head>
4   <meta charset="UTF-8" />
5   <meta name="viewport" content="width=device-width, initial-scale=1.0" />
6   <title>Latihan JavaScript – Praktikum 3</title>
7
8   <!-- Tailwind CSS (CDN terbaru) -->
9   <script src="https://cdn.tailwindcss.com"></script>
10  <script>
11    tailwind.config = {
12      theme: {
13        extend: {
14          fontFamily: {
15            sans: ["DM Sans", "sans-serif"],
16            display: ["Outfit", "sans-serif"],
17          },
18        },
19      },
20    };
21  </script>
22
23  <!-- Google Fonts -->
24  <link rel="preconnect" href="https://fonts.googleapis.com" />
25  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin />
26  <link
27    href="https://fonts.googleapis.com/css2?
family=DM+Sans:ital,opsz,wght@0,9..40,400;0,9..40,500;0,9..40,600;0,9..40,700;1,9..40,400&family
=Outfit:wght@500;600;700&display=swap"
28    rel="stylesheet"
29  />
30
31  <!-- Tabler Icons -->
32  <link
33    rel="stylesheet"
34    href="https://cdn.jsdelivr.net/npm/@tabler/icons-webfont@latest/tabler-icons.min.css"
35  />
36
37  <style>
38    body {
39      background:
40        radial-gradient(ellipse 80% 50% at 20% -10%, #dbeafe 0%, transparent 55%),
41        radial-gradient(ellipse 60% 40% at 90% 10%, #ecfcb 0%, transparent 50%),
42        #f8fafc;
43    }
44  </style>
45 </head>
46 <body class="font-sans text-slate-800 min-h-screen">
47   <div class="max-w-3xl mx-auto px-4 py-10 sm:py-14">
48     <!-- Header -->
49     <header class="mb-8">
50       <p class="text-sm font-medium text-sky-700 mb-1">Praktikum 3 · PABWE</p>
51       <h1 class="font-display text-3xl sm:text-4xl font-bold tracking-tight text-slate-900">
52         Latihan JavaScript
53       </h1>
54       <p class="mt-2 text-slate-600">
55         DOM, localStorage, logika tebak angka, dan validasi password.
56       </p>
57     </header>
58
59     <!-- Tab Navigation -->

```

```

60     <nav class="flex gap-1 p-1 bg-white/80 border border-slate-200 rounded-xl shadow-sm mb-6"
role="tablist">
61         <button
62             type="button"
63             data-tab="todo"
64             class="tab-btn flex-1 flex items-center justify-center gap-2 px-3 py-2.5 rounded-lg
text-sm font-semibold transition
65                 bg-sky-600 text-white shadow"
66             aria-selected="true"
67         >
68             <i class="ti ti-list-check text-lg"></i>
69             <span class="hidden sm:inline">Todo List</span>
70             <span class="sm:hidden">Todo</span>
71         </button>
72         <button
73             type="button"
74             data-tab="magic"
75             class="tab-btn flex-1 flex items-center justify-center gap-2 px-3 py-2.5 rounded-lg
text-sm font-semibold transition
76                 text-slate-600 hover:bg-slate-100"
77             aria-selected="false"
78         >
79             <i class="ti ti-dice-5 text-lg"></i>
80             <span class="hidden sm:inline">Magic Number</span>
81             <span class="sm:hidden">Magic</span>
82         </button>
83         <button
84             type="button"
85             data-tab="password"
86             class="tab-btn flex-1 flex items-center justify-center gap-2 px-3 py-2.5 rounded-lg
text-sm font-semibold transition
87                 text-slate-600 hover:bg-slate-100"
88             aria-selected="false"
89         >
90             <i class="ti ti-lock text-lg"></i>
91             <span class="hidden sm:inline">Password Checker</span>
92             <span class="sm:hidden">Password</span>
93         </button>
94     </nav>
95
96     <!-- ===== TAB 1: Todo List ===== -->
97     <section id="panel-todo" class="tab-panel space-y-5">
98         <div class="bg-white border border-slate-200 rounded-2xl p-5 sm:p-6 shadow-sm">
99             <h2 class="font-display text-xl font-semibold text-slate-900 mb-4 flex items-center gap-
2">
100                 <i class="ti ti-checkbox text-sky-600"></i>
101                 Todo List
102             </h2>
103
104             <!-- Form tambah todo -->
105             <form id="todo-form" class="flex flex-col sm:flex-row gap-3 mb-4">
106                 <input
107                     type="text"
108                     id="todo-title"
109                     placeholder="Judul todo..."
110                     required
111                     class="flex-1 rounded-lg border border-slate-300 px-3 py-2.5 text-sm focus:outline-
none focus:ring-2 focus:ring-sky-500 focus:border-sky-500"
112                 </>
113                 <select
114                     id="todo-status"

```

```

115         class="rounded-lg border border-slate-300 px-3 py-2.5 text-sm focus:outline-none
focus:ring-2 focus:ring-sky-500"
116     >
117         <option value="Todo">Todo</option>
118         <option value="Progress">Progress</option>
119         <option value="Done">Done</option>
120         <option value="Cancel">Cancel</option>
121     </select>
122     <button
123         type="submit"
124         class="inline-flex items-center justify-center gap-1.5 rounded-lg bg-sky-600
hover:bg-sky-700 text-white text-sm font-semibold px-4 py-2.5 transition"
125     >
126         <i class="ti ti-plus"></i>
127         Tambah
128     </button>
129 </form>
130
131 <!-- Pencarian & sorting -->
132 <div class="flex flex-col sm:flex-row gap-3 mb-4">
133     <div class="relative flex-1">
134         <i class="ti ti-search absolute left-3 top-1/2 -translate-y-1/2 text-slate-400"></i>
135         <input
136             type="search"
137             id="todo-search"
138             placeholder="Cari todo..."
139             class="w-full rounded-lg border border-slate-300 pl-9 pr-3 py-2.5 text-sm
focus:outline-none focus:ring-2 focus:ring-sky-500"
140         />
141     </div>
142     <select
143         id="todo-sort"
144         class="rounded-lg border border-slate-300 px-3 py-2.5 text-sm focus:outline-none
focus:ring-2 focus:ring-sky-500"
145     >
146         <option value="newest">Terbaru</option>
147         <option value="oldest">Terlama</option>
148         <option value="title-asc">Judul A-Z</option>
149         <option value="title-desc">Judul Z-A</option>
150         <option value="status">Status</option>
151     </select>
152 </div>
153
154 <!-- Alert jika kosong -->
155 <div
156     id="todo-empty"
157     class="hidden rounded-xl border border-amber-200 bg-amber-50 text-amber-900 px-4 py-3
text-sm flex items-start gap-2"
158     role="alert"
159 >
160     <i class="ti ti-alert-circle text-lg mt-0.5 shrink-0"></i>
161     <span>Belum ada todo. Tambahkan todo baru untuk memulai.</span>
162 </div>
163
164 <!-- Daftar todo (diisi via DOM) -->
165 <ul id="todo-list" class="space-y-2"></ul>
166 </div>
167 </section>
168
169 <!-- ===== TAB 2: Magic Number ===== -->
170 <section id="panel-magic" class="tab-panel hidden space-y-5">

```

```

171     <div class="bg-white border border-slate-200 rounded-2xl p-5 sm:p-6 shadow-sm">
172         <h2 class="font-display text-xl font-semibold text-slate-900 mb-2 flex items-center gap-
2">
173             <i class="ti ti-numbers text-lime-600"></i>
174             Magic Number
175         </h2>
176         <p class="text-sm text-slate-600 mb-5">
177             Tebak angka rahasia antara <strong>1-1000</strong>. Semakin sedikit tebakan, semakin
baik skornya.
178         </p>
179
180         <div class="grid sm:grid-cols-2 gap-4 mb-5">
181             <div class="rounded-xl bg-slate-50 border border-slate-200 px-4 py-3">
182                 <p class="text-xs uppercase tracking-wide text-slate-500 font-medium">Tebakan ke-
</p>
183                 <p id="magic-attempts" class="font-display text-2xl font-bold text-slate-900">0</p>
184             </div>
185             <div class="rounded-xl bg-lime-50 border border-lime-200 px-4 py-3">
186                 <p class="text-xs uppercase tracking-wide text-lime-700 font-medium">Top Score</p>
187                 <p id="magic-top-score" class="font-display text-2xl font-bold text-lime-800"></p>
188             </div>
189         </div>
190
191         <form id="magic-form" class="flex flex-col sm:flex-row gap-3 mb-4">
192             <input
193                 type="number"
194                 id="magic-guess"
195                 min="1"
196                 max="1000"
197                 step="1"
198                 inputmode="numeric"
199                 placeholder="Masukkan tebakan bilangan bulat (1-1000)"
200                 required
201                 class="flex-1 rounded-lg border border-slate-300 px-3 py-2.5 text-sm focus:outline-
none focus:ring-2 focus:ring-lime-500"
202             />
203             <button
204                 type="submit"
205                 class="inline-flex items-center justify-center gap-1.5 rounded-lg bg-lime-600
hover:bg-lime-700 text-white text-sm font-semibold px-4 py-2.5 transition"
206             >
207                 <i class="ti ti-target"></i>
208                 Tebak
209             </button>
210         </form>
211
212         <div
213             id="magic-feedback"
214             class="rounded-xl border border-slate-200 bg-slate-50 text-slate-700 px-4 py-3 text-sm
flex items-start gap-2"
215             role="status"
216         >
217             <i class="ti ti-info-circle text-lg mt-0.5 shrink-0"></i>
218             <span>Silakan mulai menebak!</span>
219         </div>
220
221         <button
222             type="button"
223             id="magic-reset"
224             class="mt-4 inline-flex items-center gap-1.5 text-sm font-medium text-slate-600
hover:text-slate-900 transition"

```

```

225         >
226         <i class="ti ti-refresh"></i>
227         Mulai permainan baru
228     </button>
229 </div>
230 </section>
231
232 <!-- ===== TAB 3: Password Checker ===== -->
233 <section id="panel-password" class="tab-panel hidden space-y-5">
234     <div class="bg-white border border-slate-200 rounded-2xl p-5 sm:p-6 shadow-sm">
235         <h2 class="font-display text-xl font-semibold text-slate-900 mb-2 flex items-center gap-
236 2">
237             <i class="ti ti-shield-lock text-violet-600"></i>
238             Password Checker
239         </h2>
240         <p class="text-sm text-slate-600 mb-5">
241             Kekuatan dihitung mengikuti praktik modern (panjang, keragaman karakter, dan penalti
242             pola lemah / password umum)—bukan hanya jumlah kriteria.
243         </p>
244         <div class="relative mb-4">
245             <input
246                 type="password"
247                 id="password-input"
248                 placeholder="Masukkan password..."
249                 autocomplete="new-password"
250                 class="w-full rounded-lg border border-slate-300 pl-3 pr-11 py-2.5 text-sm
251                 focus:outline-none focus:ring-2 focus:ring-violet-500"
252             />
253             <button
254                 type="button"
255                 id="password-toggle"
256                 class="absolute right-2 top-1/2 -translate-y-1/2 p-1.5 rounded-md text-slate-500
257                 hover:bg-slate-100"
258                 aria-label="Tampilkan password"
259             >
260                 <i class="ti ti-eye text-lg"></i>
261             </button>
262         </div>
263
264         <!-- Progress bar kekuatan -->
265         <div class="mb-2 flex items-center justify-between text-sm">
266             <span class="font-medium text-slate-700">Kekuatan</span>
267             <span id="password-strength-label" class="font-semibold text-slate-500">Belum
268             diisi</span>
269         </div>
270         <div class="h-2.5 rounded-full bg-slate-100 overflow-hidden mb-2">
271             <div
272                 id="password-strength-bar"
273                 class="h-full w-0 rounded-full bg-slate-300 transition-all duration-300"
274             ></div>
275         </div>
276         <p id="password-strength-hint" class="text-xs text-slate-500 mb-5"></p>
277
278         <!-- Kriteria (centang jika terpenuhi) -->
279         <ul class="space-y-2.5 id="password-criteria">
280             <li data-rule="length" class="criterion flex items-center gap-2.5 text-sm text-slate-
281 500">
282                 <i class="ti ti-circle text-base shrink-0"></i>
283                 <span>Minimal 12 karakter (disarankan standar modern)</span>
284             </li>

```

```

280         <li data-rule="upper" class="criterion flex items-center gap-2.5 text-sm text-slate-
500">
281             <i class="ti ti-circle text-base shrink-0"></i>
282             <span>Huruf besar (A-Z)</span>
283         </li>
284         <li data-rule="lower" class="criterion flex items-center gap-2.5 text-sm text-slate-
500">
285             <i class="ti ti-circle text-base shrink-0"></i>
286             <span>Huruf kecil (a-z)</span>
287         </li>
288         <li data-rule="number" class="criterion flex items-center gap-2.5 text-sm text-slate-
500">
289             <i class="ti ti-circle text-base shrink-0"></i>
290             <span>Angka (0-9)</span>
291         </li>
292         <li data-rule="special" class="criterion flex items-center gap-2.5 text-sm text-slate-
500">
293             <i class="ti ti-circle text-base shrink-0"></i>
294             <span>Karakter khusus (!@#%&^&#38;* d11.)</span>
295         </li>
296         <li data-rule="noCommon" class="criterion flex items-center gap-2.5 text-sm text-
slate-500">
297             <i class="ti ti-circle text-base shrink-0"></i>
298             <span>Bukan password umum atau pola mudah ditebak</span>
299         </li>
300     </ul>
301 </div>
302 </section>
303 </div>
304
305 <!-- Modal: Ubah Todo -->
306 <div
307     id="modal-edit"
308     class="fixed inset-0 z-50 hidden items-center justify-center p-4"
309     role="dialog"
310     aria-modal="true"
311     aria-labelledby="modal-edit-title"
312 >
313     <div class="modal-backdrop absolute inset-0 bg-slate-900/40"></div>
314     <div class="relative w-full max-w-md rounded-2xl bg-white border border-slate-200 shadow-xl
p-5 sm:p-6">
315         <div class="flex items-start justify-between gap-3 mb-4">
316             <h3 id="modal-edit-title" class="font-display text-lg font-semibold text-slate-900 flex
items-center gap-2">
317                 <i class="ti ti-pencil text-sky-600"></i>
318                 Ubah Todo
319             </h3>
320             <button type="button" data-close-modal="edit" class="p-1.5 rounded-lg text-slate-500
hover:bg-slate-100" aria-label="Tutup">
321                 <i class="ti ti-x text-lg"></i>
322             </button>
323         </div>
324
325         <form id="edit-form" class="space-y-4">
326             <div>
327                 <label for="edit-title" class="block text-sm font-medium text-slate-700 mb-
1">Judul</label>
328                 <input
329                     type="text"
330                     id="edit-title"
331                     required

```

```

332         class="w-full rounded-lg border border-slate-300 px-3 py-2.5 text-sm focus:outline-
none focus:ring-2 focus:ring-sky-500"
333     />
334 </div>
335 <div>
336     <label for="edit-status" class="block text-sm font-medium text-slate-700 mb-
1">Status</label>
337     <select
338         id="edit-status"
339         class="w-full rounded-lg border border-slate-300 px-3 py-2.5 text-sm focus:outline-
none focus:ring-2 focus:ring-sky-500"
340     >
341         <option value="Todo">Todo</option>
342         <option value="Progress">Progress</option>
343         <option value="Done">Done</option>
344         <option value="Cancel">Cancel</option>
345     </select>
346 </div>
347 <div class="flex justify-end gap-2 pt-1">
348     <button
349         type="button"
350         data-close-modal="edit"
351         class="rounded-lg border border-slate-200 px-4 py-2 text-sm font-semibold text-
slate-700 hover:bg-slate-50"
352     >
353         Batal
354     </button>
355     <button
356         type="submit"
357         class="inline-flex items-center gap-1.5 rounded-lg bg-sky-600 hover:bg-sky-700 text-
white text-sm font-semibold px-4 py-2"
358     >
359         <i class="ti ti-device-floppy"></i>
360         Simpan
361     </button>
362 </div>
363 </form>
364 </div>
365 </div>
366
367 <!-- Modal: Hapus Todo -->
368 <div
369     id="modal-delete"
370     class="fixed inset-0 z-50 hidden items-center justify-center p-4"
371     role="dialog"
372     aria-modal="true"
373     aria-labelledby="modal-delete-title"
374 >
375     <div class="modal-backdrop absolute inset-0 bg-slate-900/40"></div>
376     <div class="relative w-full max-w-md rounded-2xl bg-white border border-slate-200 shadow-xl
p-5 sm:p-6">
377         <div class="flex items-start justify-between gap-3 mb-3">
378             <h3 id="modal-delete-title" class="font-display text-lg font-semibold text-slate-900
flex items-center gap-2">
379                 <i class="ti ti-trash text-rose-600"></i>
380                 Hapus Todo
381             </h3>
382             <button type="button" data-close-modal="delete" class="p-1.5 rounded-lg text-slate-500
hover:bg-slate-100" aria-label="Tutup">
383                 <i class="ti ti-x text-lg"></i>
384             </button>

```

```

385     </div>
386
387     <p class="text-sm text-slate-600 mb-5">
388         Yakin ingin menghapus
389         <strong id="delete-todo-title" class="text-slate-900"></strong>?
390         Tindakan ini tidak dapat dibatalkan.
391     </p>
392
393     <div class="flex justify-end gap-2">
394         <button
395             type="button"
396             data-close-modal="delete"
397             class="rounded-lg border border-slate-200 px-4 py-2 text-sm font-semibold text-slate-
700 hover:bg-slate-50"
398             >
399             Batal
400         </button>
401         <button
402             type="button"
403             id="delete-confirm"
404             class="inline-flex items-center gap-1.5 rounded-lg bg-rose-600 hover:bg-rose-700 text-
white text-sm font-semibold px-4 py-2"
405             >
406             <i class="ti ti-trash"></i>
407             Hapus
408         </button>
409     </div>
410 </div>
411 </div>
412
413 <script src="assets/script.js"></script>
414 </body>
415 </html>
416

```



C. Studi Kasus

Studi kasus praktikum ini mensimulasikan pembuatan **aplikasi web interaktif satu halaman** (single page) dengan **JavaScript**. Mahasiswa diminta membangun tiga fitur berbeda dalam satu proyek, dipisah menggunakan **tab**, lalu menghubungkan perilaku UI dengan logika di **assets/script.js**.

Tema aplikasi (contoh, boleh diganti branding-nya):

- Nama: bebas (contoh: *DompétKu*, *LinkVault*, *KuisKilat*)
- Fokus: pencatatan keuangan ringan, manajemen tautan, dan kuis interaktif
- Bahasa konten UI: **Bahasa Indonesia** (kecuali istilah teknis yang umum)

Catatan umum:

- Gunakan semantic HTML5 (**header** , **nav** , **main** , **section** , **footer** , dsb.).
- Halaman harus **responsive** (terlihat layak di desktop dan mobile).
- Wajib memakai **JavaScript eksternal** (**assets/script.js**). Hindari menaruh logika

utama di atribut inline seperti `onclick="..."`.

🌸 Data yang perlu bertahan setelah refresh wajib disimpan di `localStorage`.

🌸 Diperbolehkan memakai **Tailwind CSS (CDN)**, **Google Fonts**, dan ikon (**Tabler Icons** / sejenis) untuk styling — fokus penilaian tetap pada logika JavaScript & DOM.

🌸 File latihan (`index.html` + `assets/script.js` pada proyek latihan) boleh dijadikan **referensi belajar**, bukan untuk dikumpulkan apa adanya / di-copy mentah.

Nama proyek untuk studi kasus ini menggunakan format `{username-kamu}-pabwe-p3`.

Contoh: `ifs18005-pabwe-p3`.

Proyek ini **hanya** menggunakan file `index.html` dan `assets/script.js` (aset pendukung opsional: gambar di `assets/img/`). Pisahkan ketiga fitur menggunakan **tab**.

🌻 3.1 Catatan Pengeluaran Harian (Expense Tracker)

Tujuan: membuat fitur pencatatan transaksi harian dengan **CRUD via DOM**, perhitungan ringkas, pencarian/filter, dan penyimpanan di `localStorage`.

Nama file (dalam proyek yang sama):

🌸 `index.html` — markup tab + panel Expense Tracker

🌸 `assets/script.js` — seluruh logika JavaScript

Catatan:

🌸 Setiap transaksi minimal memiliki: **judul/deskripsi**, **kategori**, **jumlah (angka)**, **tipe** (`Pemasukan` / `Pengeluaran`), dan **tanggal**.

🌸 Tampilkan **ringkasan**: total pemasukan, total pengeluaran, dan saldo.

🌸 Ubah dan hapus sebaiknya memakai **modal** (bukan `prompt` / `confirm` browser saja).

🌸 Validasi input: jumlah harus angka valid dan lebih dari 0; field wajib tidak boleh kosong.

🌸 Jika belum ada data, tampilkan **pesan kosong** (alert / empty state).

Hints — perilaku minimal:

1. Form tambah transaksi

🌸 Input judul, kategori (select atau text), jumlah, tipe, tanggal

🌸 Submit menambah item ke array lalu render ulang daftar

2. Daftar transaksi

🌸 Ditampilkan lewat manipulasi DOM (`createElement` / update `innerHTML`)

yang terkendali)

🌸 Badge tipe / kategori

🌸 Tombol **Ubah** dan **Hapus**

3. Pencarian & filter

🌸 Cari berdasarkan judul

🌸 Filter berdasarkan tipe dan/atau kategori

4. Sorting (minimal salah satu): terbaru, terlama, jumlah terbesar/terkecil

5. Persistensi

🌸 Simpan array transaksi ke `localStorage`

🌸 Muat ulang data saat halaman dibuka

Hints — materi JS yang diharapkan muncul:

🌸 Selektor DOM (`querySelector` / `querySelectorAll`)

🌸 Event (`submit` , `click` , `input` / `change`)

🌸 Array & object (`push` , `filter` , `find` , `map` / `forEach` , `sort`)

🌸 `JSON.stringify` / `JSON.parse` + `localStorage`

🌸 Kondisi & validasi form

Hasil diharapkan:

Panel Expense Tracker yang dapat menambah, mengubah, menghapus, mencari/filter transaksi, menampilkan ringkasan saldo, dan datanya tetap ada setelah halaman di-refresh.

🌻 3.2 Bookmark / Link Manager

Tujuan: membuat fitur penyimpanan tautan favorit dengan validasi URL, kategori/tag, pencarian/sorting, dan `localStorage`.

Nama file (dalam proyek yang sama):

🌸 `index.html` — markup tab + panel Bookmark

🌸 `assets/script.js` — logika Bookmark (boleh dalam file yang sama, dipisah per bagian dengan komentar)

Catatan:

🌸 Setiap bookmark minimal memiliki: **nama/judul**, **URL**, **kategori/tag**, dan **catatan singkat** (opsional).

🌸 Validasi URL sederhana (minimal diawali `http://` atau `https://`).

🌸 Klik judul/URL harus dapat membuka tautan di tab baru (`target="_blank"`) +

`rel="noopener noreferrer")`.

🌸 Ubah/hapus memakai modal (konsisten dengan fitur lain).

🌸 Jika daftar kosong, tampilkan empty state.

Hints — perilaku minimal:

1. Form tambah bookmark

🌸 Nama, URL, kategori, catatan

2. Daftar bookmark

🌸 Kartu/baris berisi judul, URL (link), kategori, aksi ubah/hapus

3. Pencarian & sorting

🌸 Cari nama / URL / kategori

🌸 Sort judul A–Z / Z–A / terbaru

4. Persistensi

🌸 Simpan ke `localStorage` dengan key yang berbeda dari Expense Tracker

Hints — materi JS yang diharapkan muncul:

🌸 Validasi string / regex sederhana untuk URL

🌸 Manipulasi DOM untuk list dinamis

🌸 Event pada form dan tombol aksi

🌸 Pemisahan key `localStorage` antar fitur agar data tidak bentrok

Hasil diharapkan:

Panel Bookmark yang rapi, URL tervalidasi, tautan bisa dibuka, CRUD berfungsi, serta data tersimpan di browser.

🌻 3.3 Kuis Interaktif (Quiz App)

Tujuan: membuat kuis pilihan ganda berbasis data di JavaScript (array of object), dengan skor, feedback, dan high score di `localStorage`.

Nama file (dalam proyek yang sama):

🌸 `index.html` — markup tab + panel Kuis

🌸 `assets/script.js` — logika kuis

Catatan:

🌸 Soal disimpan sebagai **array of object** di JavaScript (bukan hardcode HTML satu per satu untuk semua soal).

🌸 Minimal **5 soal**, masing-masing punya: pertanyaan, opsi (minimal 4), dan jawaban benar.

🌸 Topik soal bebas (contoh: HTML/CSS/JS dasar, literasi digital, atau tema kampus).

🌸 Tampilkan skor akhir dan opsi **ulangi kuis**.

🌸 Simpan **skor terbaik (high score)** ke `localStorage`.

Hints — perilaku minimal:

1. Mulai kuis

🌸 Tombol mulai / reset state (nomor soal, skor, jawaban user)

2. Tampilan soal

🌸 Render pertanyaan + opsi lewat DOM

🌸 Navigasi soal berikutnya setelah menjawab (atau tombol Next)

3. Penilaian

🌸 Bandingkan jawaban user dengan kunci

🌸 Update skor

🌸 Feedback singkat (benar/salah) — opsional per soal atau di akhir

4. Hasil akhir

🌸 Tampilkan skor (misalnya `4 / 5`)

🌸 Bandingkan / update high score

🌸 Tombol main lagi

5. Bonus (opsional): timer per soal atau total waktu memakai `setInterval` / `setTimeout`

Hints — materi JS yang diharapkan muncul:

🌸 Struktur data (array of object)

🌸 State application (index soal aktif, skor, status selesai)

🌸 Render ulang UI berdasarkan state

🌸 `Math` / percabangan untuk penilaian

🌸 `localStorage` untuk high score

Hasil diharapkan:

Panel kuis yang bisa dijalankan end-to-end: jawab soal → lihat skor → high score tersimpan → bisa diulang.

🌻 3.4 Integrasi Tab & Proyek

Catatan: gabungkan ketiga fitur studi kasus dalam satu halaman dengan navigasi **tab**.

Panel yang harus terhubung:

Panel / Tab	Teknologi utama	Persistensi
Catatan Pengeluaran Harian	DOM, event, array/object, validasi angka	<code>localStorage</code> transaksi
Bookmark / Link Manager	DOM, validasi URL, CRUD	<code>localStorage</code> bookmark
Kuis Interaktif	state, render soal, skor	<code>localStorage</code> high score

Kriteria proyek:

1. Terdapat **3 tab** yang berfungsi; hanya satu panel aktif pada satu waktu.
2. Tab terakhir yang dibuka **diingat** (simpan ke `localStorage`) dan dipulihkan saat reload.
3. Setiap fitur memakai **key `localStorage` berbeda** agar data tidak saling menimpa.
4. Kode di `assets/script.js` relatif rapi, dikelompokkan per fitur, dan diberi **komentar singkat** pada bagian penting.
5. UI konsisten antar tab (warna, font, spacing) meskipun konten berbeda.
6. Tidak bergantung pada backend / API (`fetch` tidak wajib pada praktikum ini).

Struktur folder yang disarankan:

Coding Time

```

1 | nama-repo/
2 |   └─ index.html
3 |   └─ assets/
4 |       └─ script.js
5 |       └─ img/           # opsional
6 |   └─ README.md         # opsional: nama, NIM, deskripsi singkat

```

Checklist pengumpulan:

- ☐  Proyek bernama sesuai format `{username}-pabwe-p3`
- ☐  Hanya `index.html` + `assets/script.js` sebagai inti (CDN styling boleh)
- ☐  Tiga tab berjalan: Expense Tracker, Bookmark Manager, Quiz App
- ☐  Expense: CRUD + ringkasan saldo + cari/filter + `localStorage`
- ☐  Bookmark: CRUD + validasi URL + cari/sort + `localStorage`
- ☐  Quiz: ≥5 soal dari array object + skor + high score `localStorage`
- ☐  Tab terakhir diingat setelah refresh
- ☐  Ubah/hapus memakai modal (minimal pada fitur yang punya CRUD)
- ☐  Responsive diuji di browser (desktop & mobile / DevTools)
- ☐  Kode tidak hanya copy-paste mentah dari file latihan